Zhuk's Centers and Ternary Absorption

Abstract

Let $\mathbf{A}$ and $\mathbf{B}$ be finite idempotent algebras of a common signature and let R be a subdirect subuniverse of $\mathbf{A} \times \mathbf{B}$. The *left center* of R is the set C of elements of A related by R to every element of B. We give a complete and self-contained proof of Zhuk's theorem that if $\mathbf{B}$ has a Taylor term and has no nonempty proper binary absorbing subuniverse, then C absorbs $\mathbf{A}$ — and, through the Zhuk–Kozik essential doubling trick, that the absorption is witnessed by a *ternary* term. Everything is proved from the definitions of terms, substitution, subuniverses, and generation: the relational description of absorption by essential relations, which supplies the passage from unbounded arity to ternary arity, is proved here rather than imported. The background package of Appendix C is finite combinatorics and elementary set theory only. The document is written to serve as the human-readable blueprint for a formal development.

Theorem 0.1 (Main theorem)**.** *Let σ be a signature without nullary symbols and let $\mathbf{A}, \mathbf{B}$ be finite idempotent σ-algebras with nonempty universes. Let $R \leq \mathbf{A} \times \mathbf{B}$ be subdirect and let*

$$C = \{a \in A \ : \ \{a\} \times B \subseteq R\}$$

be its left center. Assume

(a) *some $t \in T_\sigma(k)$ is Taylor on B* (Definition 2.5)*, and*
(b) *$\mathbf{B}$ has no nonempty proper binary absorbing subuniverse* (Definition 2.1)*.*

Then C centrally absorbs $\mathbf{A}$, and there is a ternary *term $s \in T_\sigma(3)$ such that*

$$s^{\mathbf{A}}(C, A, C) \cup s^{\mathbf{A}}(C, C, A) \cup s^{\mathbf{A}}(A, C, C) \subseteq C.$$

Contents

Introduction

The algebraic approach to the constraint satisfaction problem studies a finite relational template through the algebra of its polymorphisms. Zhuk's proof of the dichotomy conjecture [4] isolated a mechanism that turns a single binary relation between two algebras into an absorbing subalgebra of one of them. Barto and Kozik's absorption theorem [1] is the older tool it replaces; Brady's notes [3], whose Section 3.10 this document follows, use Zhuk's mechanism to reprove and strengthen it.

Three distinct ingredients are combined here, and it is worth separating them. The left-center argument of Sections 5 and 6 is due to Zhuk [4]. The essential doubling trick of Section 7 is attributed in [3] to Zhuk and Kozik. The relational description of absorption proved in Part II, which is what converts absorption at some arity into absorption at arity 3, is due to Barto and Kazda [2]; it is a large and logically independent component of this document, and it is proved here rather than cited precisely because the ternary conclusion rests on it.

The mechanism is this. Given a subdirect $R \leq \mathbf{A} \times \mathbf{B}$, call $C = \{a : \{a\} \times B \subseteq R\}$ the *left center*. If C is neither empty nor all of A, then C is a nontrivial obstruction inside $\mathbf{A}$, and one expects to convert it into structure. Zhuk's observation is that the conversion is forced: applying a Taylor term of $\mathbf{B}$ to one element of A and several elements of C strictly enlarges its R-neighborhood in B, unless that neighborhood is a binary absorbing subalgebra of $\mathbf{B}$. Iterating the enlargement $|B| - 1$ times drives the value into C. The witnessing term has arity $k^{|B|-1}$. The second half of the argument does not compress that term: the Zhuk–Kozik doubling trick shows instead that no ternary essential relation can exist, and the relational description of absorption then yields the *existence* of some ternary witness.

What is proved, and from what. The document is self-contained relative to Appendix C, which asks only for finite sets, induction, and elementary reasoning about functions and products. In particular the following are proved rather than assumed:

- that term operations preserve subuniverses, and the description of a generated subuniverse as the set of values of terms on a fixed list of generators (Section 1);
- the relational description of absorption — B absorbs $\mathbf{A}$ with respect to an m-ary term precisely when no m-ary B-essential relation exists (Theorem 3.11) — which is the bridge between "absorbs at some arity" and "absorbs at arity 3";
- Zhuk's two theorems on left centers, the definition and basic theory of central absorption, the doubling trick, and the ternary conclusion (Section 4);
- simultaneous substitution of terms into terms, with its evaluation law (Definition 1.10, Lemma 1.11), which the constructions of Parts I and III require.

Route. Part I fixes the universal-algebraic vocabulary and proves the handful of closure facts used everywhere: subuniverses are closed under term operations, generated subuniverses are term images of a fixed generator list, homomorphisms commute with generation, and the five relational constructions of Lemma 1.19 — intersection, intersection with a box, projection, cylinder and reindexing — preserve subuniverses. All five are used: reindexing in the base case of Lemma 3.8, and the cylinder in Step 4 of Lemma 7.1. Part I ends with absorption, binary absorption, and the lemma — Lemma 2.7 — that converts a *one-sided* closure condition into binary absorption using a single Taylor identity.

Part II develops essential relations. An m-ary B-essential relation is a relation that meets every box with one coordinate free and all others confined to B, but misses the all-B box; it is exactly the obstruction to m-ary absorption. The easy direction (absorption forbids essential relations) is a one-line application of the witnessing term to the witnessing tuples. The hard direction runs the free algebra $\mathrm{Clo}_m(\mathbf{A})$ against a grouped form of the definition, and needs the regrouping Lemma 3.8.

Part III is Section 3.10 of [3]. Theorem 5.1 is the enlargement step, Theorem 5.2 its iteration, Theorem 6.1 the extra "central" property of the left center, Lemma 7.1 the doubling trick, and Corollary 8.1 the passage to arity 3.

Divergences from the source. Three points where this document is deliberately more explicit than [3]. First, the empty set is a subuniverse of every algebra in a signature without nullary symbols, and it absorbs vacuously; the hypothesis "$\mathbf{B}$ has no proper binary absorbing subalgebra" must therefore be read as excluding only *nonempty* proper ones, which is what Convention 1.5 records. Second, Theorem 6.1 is stated with the three generator blocks $\{a\} \times C$, $C \times C$, $C \times \{a\}$; the middle block is genuinely present in the source and is dropped only when passing to the weaker Definition 6.2. Third, the arities produced by iterating the doubling trick from arity 3 are $2 + 2^k$, and the argument in Corollary 8.1 is arranged here to need only that the arity strictly increases.

What the formalizations found. This document has been formalized twice, in two independent proof assistants: in Lean 4 with Mathlib, and in a second system. Both formalizations are complete and follow the numbering used here. In three places they showed that a hypothesis is redundant, an auxiliary lemma unnecessary, or a statement more general than its only consumer requires. Nothing below is incorrect, so the text is left as it stands and each point is recorded where it arises: Remark 3.3, Remark 1.21 and Remark 7.3. What is worth recording about all three is how they were found — by the two formalizations *agreeing* on a simplification, rather than by either one alone.

Part I
Algebras, terms, and absorption

1 Signatures, algebras, and subuniverses

Definition 1.1 (Signature and algebra)**.** A *signature* σ is a set of operation symbols together with an arity function $\mathrm{ar} : \sigma \to \{1, 2, 3, \dots\}$. Nullary symbols are not permitted. A σ-*algebra* $\mathbf{A}$ consists of a set A, its *universe*, together with an operation $f^{\mathbf{A}} : A^{\mathrm{ar}(f)} \to A$ for each $f \in \sigma$. The algebra is *finite* if A is finite. We write $\mathbf{A}$ for the algebra and A for its universe, and never identify the two.

Convention 1.2 (Standing hypotheses)**.** Two hypotheses are in force throughout and are not repeated in statements.

(a) All algebras are σ-algebras for one fixed signature σ, which has no nullary symbols. This is what makes $\varnothing$ a subuniverse, hence what Lemma 1.6(d) and Convention 1.5 rest on, and it is used again in Theorem 6.1.

(b) Every algebra named has nonempty universe. This is consumed in three places: Remark 2.3, where $t^{\mathbf{A}}(z_1) \in \varnothing$ must fail for some z_1; Theorem 3.11, where $R = A$ is a nonempty $\varnothing$-essential relation of arity 1; and Remark 4.4(b), where $B \neq \varnothing$ is what makes a constant idempotent unary operation force $|B| = 1$.

A formalization should carry both as explicit hypotheses or instance arguments rather than as ambient prose; the audit above records where each is actually needed.

Definition 1.3 (Idempotence)**.** $\mathbf{A}$ is *idempotent* if $f^{\mathbf{A}}(a, a, \dots, a) = a$ for every $f \in \sigma$ and every $a \in A$.

Definition 1.4 (Subuniverse)**.** A set $S \subseteq A$ is a *subuniverse* of $\mathbf{A}$, written $S \leq \mathbf{A}$, if $f^{\mathbf{A}}(s_1, \dots, s_{\mathrm{ar}(f)}) \in S$ for every $f \in \sigma$ and all $s_i \in S$. A subuniverse is *proper* if it is not equal to A.

Because σ has no nullary symbols, $\varnothing \leq \mathbf{A}$ always. This is convenient in Part II, where projections of relations may be empty, but it makes the phrase "proper absorbing subalgebra" ambiguous unless nonemptiness is legislated. We legislate it once:

Convention 1.5 (Subalgebra means nonempty)**.** A *subalgebra* of $\mathbf{A}$ is a nonempty subuniverse of $\mathbf{A}$. Whenever a hypothesis or conclusion says that an algebra *has* or *has no* proper subalgebra with some property, "subalgebra" carries its nonemptiness. Objects introduced as "subuniverses" may be empty unless stated otherwise.

Lemma 1.6 (Intersections and generation)**.** *Let $\mathbf{A}$ be an algebra.*

(a) *The intersection of any nonempty family of subuniverses of $\mathbf{A}$ is a subuniverse of $\mathbf{A}$.*
(b) *For $S \subseteq A$ there is a least subuniverse of $\mathbf{A}$ containing S, namely the intersection of all subuniverses containing S. We write it $\mathrm{Sg}_{\mathbf{A}}(S)$.*
(c) *$S \subseteq T \leq \mathbf{A}$ implies $\mathrm{Sg}_{\mathbf{A}}(S) \subseteq T$; and $S \subseteq S'$ implies $\mathrm{Sg}_{\mathbf{A}}(S) \subseteq \mathrm{Sg}_{\mathbf{A}}(S')$.*
(d) *$\mathrm{Sg}_{\mathbf{A}}(\varnothing) = \varnothing$.*

Proof. For (a), if $s_1, \dots, s_n$ lie in every member of the family then so does $f^{\mathbf{A}}(s_1, \dots, s_n)$. For (b), the family of subuniverses containing S is nonempty because $A \leq \mathbf{A}$, so (a) applies; the intersection contains S and is contained in every subuniverse containing S. Part (c) is immediate from the description in (b). For (d), $\varnothing \leq \mathbf{A}$ because σ has no nullary symbols, so (c) applies with $T = \varnothing$. □

Lemma 1.7 (Singletons)**.** *If $\mathbf{A}$ is idempotent then $\{a\} \leq \mathbf{A}$ for every $a \in A$.*

Proof. $f^{\mathbf{A}}(a, \dots, a) = a$ by Definition 1.3. □

Definition 1.8 (Products)**.** Given a family of algebras $(\mathbf{A}_i)_{i \in I}$ indexed by a *finite* set I, the product algebra $\prod_{i \in I} \mathbf{A}_i$ has universe $\prod_{i \in I} A_i$ and operations applied coordinatewise. A *relation* is a subuniverse $R \leq \prod_{i \in I} \mathbf{A}_i$; when $I = [m]$ we write $\mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and call m the *arity*. We write $\mathbf{A}^m$ for the m-fold product of $\mathbf{A}$ with itself, and $\mathbf{A}^X$ for the product of copies of $\mathbf{A}$ indexed by a set X, whose universe is the set of functions $X \to A$.

The index set is deliberately allowed to be an arbitrary finite set rather than an initial segment $[m]$: Theorem 3.11 forms $\mathbf{A}^{A^m}$ and projects it onto a subset of A^m. Finiteness is retained because a product of nonempty universes over an infinite index set is nonempty only by a choice principle, and every index set used below is finite — A^m and its subsets because A is finite, and the products of Lemma 7.1 because they have finitely many factors. Keeping I finite therefore holds the foundational boundary of Appendix C at finite choice.

Definition 1.9 (Terms)**.** Let V be a nonempty set, whose members we call *variables.* The set $T_\sigma(V)$ of *terms over* V is defined inductively: for each $v \in V$ the variable x_v is a term, and if $f \in \sigma$ has arity k and $t_1, \dots, t_k \in T_\sigma(V)$ then $f(t_1, \dots, t_k) \in T_\sigma(V)$. For an algebra $\mathbf{A}$, the *term operation* $t^{\mathbf{A}} : A^V \to A$ is defined by $x_v^{\mathbf{A}}(\vec{a}) = a_v$ and $f(t_1, \dots, t_k)^{\mathbf{A}}(\vec{a}) = f^{\mathbf{A}}(t_1^{\mathbf{A}}(\vec{a}), \dots, t_k^{\mathbf{A}}(\vec{a}))$.

For $n \geq 1$ we abbreviate $T_\sigma(n) := T_\sigma([n])$, write its variables as $x_1, \dots, x_n$, call its members *n-ary*, and read $t^{\mathbf{A}}$ as a map $A^n \to A$; we write $\mathrm{Clo}_n(\mathbf{A}) = \{t^{\mathbf{A}} : t \in T_\sigma(n)\} \subseteq A^{A^n}$. Nearly every term below is n-ary. The general variable set earns its place in exactly one construction, Definition 2.8, where the natural index set for a star power is a set of *functions* rather than an initial segment; see the remark following that definition.

Definition 1.10 (Simultaneous substitution)**.** Let V, W be nonempty variable sets. For $t \in T_\sigma(V)$ and a family $u = (u_v)_{v \in V}$ with $u_v \in T_\sigma(W)$, the term $t[u] \in T_\sigma(W)$ is defined by structural recursion on t:

$$x_v[u] := u_v, \qquad f(t_1, \dots, t_k)[u] := f(t_1[u], \dots, t_k[u]).$$

When $V = [m]$ we write the family as a list and the term as $t[u_1, \dots, u_m]$. For an arbitrary map $\rho : V \to W$ we abbreviate $t[\rho] := t[(x_{\rho(v)})_{v \in V}] \in T_\sigma(W)$. Nothing is required of ρ: permutation of variables, identification of variables and introduction of dummy variables are all covered, and the construction is single-valued because it substitutes *into* variables rather than searching for preimages.

Lemma 1.11 (Evaluation of a substitution)**.** *Let $t \in T_\sigma(V)$, let $u = (u_v)_{v \in V}$ with $u_v \in T_\sigma(W)$, let $\mathbf{A}$ be an algebra and let $\vec{a} \in A^W$. Then*

$$(t[u])^{\mathbf{A}}(\vec{a}) = t^{\mathbf{A}}(u^{\mathbf{A}}(\vec{a})), \qquad \textit{where } u^{\mathbf{A}}(\vec{a}) \in A^V \textit{ is the tuple } v \mapsto u_v^{\mathbf{A}}(\vec{a}).$$

In particular, for $\rho : V \to W$, $(t[\rho])^{\mathbf{A}}(\vec{a}) = t^{\mathbf{A}}(\vec{a} \circ \rho)$; with $V = [m]$ and $W = [n]$ this reads $(t[\rho])^{\mathbf{A}}(a_1, \dots, a_n) = t^{\mathbf{A}}(a_{\rho(1)}, \dots, a_{\rho(m)})$.

Proof. Structural induction on t. If $t = x_v$, both sides equal $u_v^{\mathbf{A}}(\vec{a})$. If $t = f(t_1, \dots, t_k)$, then Definition 1.9 and the inductive hypothesis give

$$(f(t_1, \dots, t_k)[u])^{\mathbf{A}}(\vec{a}) = f^{\mathbf{A}}\Big((t_1[u])^{\mathbf{A}}(\vec{a}), \dots\Big) = f^{\mathbf{A}}\Big(t_1^{\mathbf{A}}(u^{\mathbf{A}}(\vec{a})), \dots\Big) = t^{\mathbf{A}}(u^{\mathbf{A}}(\vec{a})).$$

The special case follows because $x_{\rho(v)}^{\mathbf{A}}(\vec{a}) = a_{\rho(v)}$. ☐

Remark 1.12 (Where substitution is used)**.** Definition 1.10 is not decoration, and it is not merely renaming: the binary term of Lemma 2.7 is $t[\rho]$ for a map $\rho : [k] \to [2]$ that *identifies* variables; the star powers of Definition 2.8 substitute terms into a term; and the binary term at the end of Theorem 6.1 is $t[\rho]$ for a map $\rho : [n] \to [2]$. Renaming along a bijection is used once, in Lemma 2.2, and only to convert a witness over an arbitrary finite variable set into one of numeric arity. Substitution therefore belongs to the foundations of Part I, not to the imported background of Appendix C.

Lemma 1.13 (Term operations are idempotent)**.** *If $\mathbf{A}$ is idempotent then $t^{\mathbf{A}}(\vec{a}) = a$ for every nonempty variable set V, every $t \in T_\sigma(V)$ and every constant tuple $\vec{a} \in A^V$ with all coordinates equal to $a \in A$.*

Proof. Induction on t. For a variable the value is a. For $f(t_1, \dots, t_k)$ the value is $f^{\mathbf{A}}(a, \dots, a) = a$ by the inductive hypothesis and Definition 1.3. ☐

Lemma 1.14 (Preservation)**.** *Let $S \leq \mathbf{A}$, let V be a nonempty variable set and let $t \in T_\sigma(V)$. Then $t^{\mathbf{A}}(\vec{s}) \in S$ for every $\vec{s} \in A^V$ with $s_v \in S$ for all $v \in V$.*

Proof. Induction on t. A variable returns one of the s_v. For $f(t_1, \dots, t_k)$, each $t_j^{\mathbf{A}}(\vec{s}) \in S$ by the inductive hypothesis, and S is closed under $f^{\mathbf{A}}$ by Definition 1.4. ☐

Applied to a product algebra, Lemma 1.14 is the statement that relations are *preserved* by term operations, in the form used constantly below: if $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ and $(a_{1j}, \dots, a_{mj}) \in R$ for $j = 1, \dots, n$, then

$$(t^{\mathbf{A}_1}(a_{11}, \dots, a_{1n}), \ \dots, \ t^{\mathbf{A}_m}(a_{m1}, \dots, a_{mn})) \in R.$$

Lemma 1.15 (Generation by a fixed list)**.** *Let* $\mathbf{A}$ *be an algebra and let* $S = \{s_1, \dots, s_N\} \subseteq A$ *with* $N \geq 1$. *Then*

$$\mathrm{Sg}_{\mathbf{A}}(S) = \{\, t^{\mathbf{A}}(s_1, \dots, s_N) \; : \; t \in T_\sigma(N) \,\}.$$

Proof. Write E for the right-hand side. Every element of E lies in $\mathrm{Sg}_{\mathbf{A}}(S)$ by Lemma 1.14, since $S \subseteq \mathrm{Sg}_{\mathbf{A}}(S) \leq \mathbf{A}$. Conversely E contains each s_i, taking $t = x_i$; and $E \leq \mathbf{A}$, because for $f \in \sigma$ of arity k and $t_1, \dots, t_k \in T_\sigma(N)$,

$$f^{\mathbf{A}}\big(t_1^{\mathbf{A}}(\vec{s}), \dots, t_k^{\mathbf{A}}(\vec{s})\big) = \big(f(t_1, \dots, t_k)\big)^{\mathbf{A}}(\vec{s}) \in E.$$

Hence $\mathrm{Sg}_{\mathbf{A}}(S) \subseteq E$ by Lemma 1.6(c). □

Remark 1.16 (Why a fixed list)**.** The usual formulation — every element of $\mathrm{Sg}_{\mathbf{A}}(S)$ is $t^{\mathbf{A}}$ applied to *some* finite tuple of generators — forces one to combine terms written over different generator lists, and so to reason about arbitrary assignments of generators to variables. Fixing the list once, which is possible because every generating set used below is finite, removes that bookkeeping: each element of $\mathrm{Sg}_{\mathbf{A}}(S)$ is named by a single term of one fixed arity. This is a simplification of the *indexing*, not an elimination of substitution; substitution is used freely elsewhere, as Remark 1.12 records.

Definition 1.17 (Homomorphism)**.** A map $h : A \to B$ is a *homomorphism* $\mathbf{A} \to \mathbf{B}$ if $h(f^{\mathbf{A}}(a_1, \dots, a_k)) = f^{\mathbf{B}}(h(a_1), \dots, h(a_k))$ for all $f \in \sigma$ and $a_i \in A$. Each coordinate projection $\pi_i : \mathbf{A}_1 \times \dots \times \mathbf{A}_m \to \mathbf{A}_i$, and more generally each projection π_J onto a subfamily of coordinates, is a homomorphism.

Lemma 1.18 (Images and generation)**.** *Let* $h : \mathbf{A} \to \mathbf{B}$ *be a homomorphism.*

(a) *If* $S \leq \mathbf{A}$ *then* $h(S) \leq \mathbf{B}$.
(b) $h(t^{\mathbf{A}}(a_1, \dots, a_n)) = t^{\mathbf{B}}(h(a_1), \dots, h(a_n))$ *for every* $t \in T_\sigma(n)$.
(c) $h(\mathrm{Sg}_{\mathbf{A}}(S)) = \mathrm{Sg}_{\mathbf{B}}(h(S))$ *for every finite nonempty* $S \subseteq A$.

Proof. (a) An element of $h(S)$ has the form $h(s)$; apply the homomorphism identity and closure of S. (b) Induction on t, using the homomorphism identity at each operation symbol. (c) Enumerate $S = \{s_1, \dots, s_N\}$; by Lemma 1.15 both sides equal $\{t^{\mathbf{B}}(h(s_1), \dots, h(s_N)) : t \in T_\sigma(N)\}$, the left-hand side by (b). □

Lemma 1.19 (Relational constructions)**.** *Let* $(\mathbf{A}_i)_{i \in I}$ *be a family of algebras indexed by a finite set* I.

(a) *If* $R, R' \leq \prod_{i \in I} \mathbf{A}_i$ *then* $R \cap R' \leq \prod_{i \in I} \mathbf{A}_i$.
(b) *If* $R \leq \prod_{i \in I} \mathbf{A}_i$ *and* $S_i \leq \mathbf{A}_i$ *for each* i, *then* $R \cap \prod_{i \in I} S_i \leq \prod_{i \in I} \mathbf{A}_i$.
(c) *If* $R \leq \prod_{i \in I} \mathbf{A}_i$ *and* $J \subseteq I$ *then* $\pi_J(R) \leq \prod_{i \in J} \mathbf{A}_i$.
(d) *If* $h : \mathbf{A} \to \mathbf{B}$ *is a homomorphism and* $S \leq \mathbf{B}$, *then* $h^{-1}(S) \leq \mathbf{A}$. *In particular, for* $J \subseteq I$ *and* $S \leq \prod_{i \in J} \mathbf{A}_i$, *the cylinder* $\pi_J^{-1}(S)$ *is a subuniverse of* $\prod_{i \in I} \mathbf{A}_i$.
(e) *If* $\rho : I' \to I$ *is a bijection, the induced map* $(a_i)_{i \in I} \mapsto (a_{\rho(i')})_{i' \in I'}$ *is an isomorphism* $\prod_{i \in I} \mathbf{A}_i \to \prod_{i' \in I'} \mathbf{A}_{\rho(i')}$, *and carries subuniverses to subuniverses.*

Proof. (a) is Lemma 1.6(a). (b) The box $\prod_{i \in I} S_i$ is a subuniverse because operations act coordinatewise, so (a) applies. (c) is Lemma 1.18(a) applied to the projection homomorphism. (d) If $h(a_r) \in S$ for $r \in [k]$ then $h(f^{\mathbf{A}}(a_1, \dots, a_k)) = f^{\mathbf{B}}(h(a_1), \dots, h(a_k)) \in S$. (e) The map is a bijection commuting with coordinatewise operations, so Lemma 1.18(a) applies. Nothing in (a)–(e) uses an ordering of I. □

Lemma 1.20 (Block-respecting enumeration)**.** *Let I be a finite set partitioned into blocks $I_1, \dots, I_m$, put $n_j := |I_j|$ and $n := |I|$. Then there is an enumeration $i_1, \dots, i_n$ of I listing the blocks consecutively in order: the first n_1 terms are the elements of I_1, the next n_2 are those of I_2, and so on.*

Proof. Enumerate each block separately, which is possible because each is finite, and concatenate the enumerations. The result lists each element exactly once because the blocks partition I. □

Remark 1.21 (The enumeration is avoidable)**.** Lemma 1.20 has one consumer: the proof of Theorem 6.1 enumerates the generators so that the three blocks appear consecutively, which is what lets it name the indices $p \leq q$ and choose m between them. Neither formalization needed it. Both take the variable type of the term supplied by Lemma 1.15 to *be* the generating set, so each variable already records which block it came from, no sorted enumeration is required, and the selector μ of that proof reduces to the test "is this generator's second coordinate a". The lemma is retained because the proof as written here uses it.

Remark 1.22 (The only relational constructions used)**.** Lemma 1.19 contains all the relational closure facts used in this document; no general theory of primitive positive definability is required. Its principal applications are the arity-reduction and regrouping constructions of Part II, the neighborhoods $a + R$ and the right center of Part III, and the relations $\beta(R)$, P, Q and R' built in Lemma 7.1. All five parts are needed: reindexing appears in the base case of Lemma 3.8 and in the right center, and the cylinder appears in Step 4 of Lemma 7.1.

2 Absorption

Definition 2.1 (Absorption)**.** Let $\mathbf{A}$ be an algebra, let $E \subseteq D$ be subuniverses of $\mathbf{A}$ and let V be a finite nonempty variable set. For $t \in T_\sigma(V)$ we say that t *witnesses* $E \lhd D$ if for every $i \in V$ and every $\vec{z} \in A^V$,

$$t^{\mathbf{A}}(\vec{z}) \in E \qquad \text{whenever } z_i \in D \text{ and } z_j \in E \text{ for all } j \neq i.$$

For $V = [m]$ this reads: for every $i \in [m]$, $t^{\mathbf{A}}(z_1, \dots, z_m) \in E$ whenever $z_i \in D$ and $z_j \in E$ for all $j \neq i$.

We write $E \lhd D$, and say E *absorbs* D, if some $t \in T_\sigma(V)$ witnesses it for some finite nonempty V — equivalently, by Lemma 2.2, if some $t \in T_\sigma(m)$ witnesses it for some $m \geq 1$. We write $E \lhd_{\text{bin}} D$, and say E *binary absorbs* D, if some $t \in T_\sigma(2)$ witnesses it, i.e. if $t^{\mathbf{A}}(E \times D) \cup t^{\mathbf{A}}(D \times E) \subseteq E$. When $D = A$ we also write $E \lhd \mathbf{A}$ and $E \lhd_{\text{bin}} \mathbf{A}$.

The ambient algebra is suppressed in the notation. This is harmless: the condition mentions $t^{\mathbf{A}}$ only at arguments lying in D, so it depends on $\mathbf{A}$ only through the operations of $\mathbf{A}$ restricted to D. No separate notion of "the subalgebra on D" is needed.

The variable set is allowed to be an arbitrary finite set for the same reason the index set of a product is (Definition 1.8): the terms constructed in Definition 2.8 are naturally indexed by a set of functions, and forcing them onto $[k^\ell]$ would require an explicit Euclidean-division bijection at every step of the induction in Theorem 5.2. The following lemma is what makes the generality free — it is used only to return to a numeric arity at the end.

Lemma 2.2 (Renaming the variable set)**.** *Let $E \subseteq D$ be subuniverses of $\mathbf{A}$, let $\beta : V \to W$ be a bijection of nonempty variable sets, and let $t \in T_\sigma(V)$ witness $E \lhd D$. Then $t[\beta] \in T_\sigma(W)$ witnesses $E \lhd D$. In particular, if V is finite with $|V| = m$, then some term of $T_\sigma(m)$ witnesses $E \lhd D$.*

Proof. Let $i \in W$ and let $\vec{a} \in A^W$ satisfy $a_i \in D$ and $a_j \in E$ for all $j \neq i$. By Lemma 1.11, $(t[\beta])^{\mathbf{A}}(\vec{a}) = t^{\mathbf{A}}(\vec{a} \circ \beta)$. Put $i' := \beta^{-1}(i)$. Then $(\vec{a} \circ \beta)_{i'} = a_i \in D$, and for every $j' \neq i'$ we have $\beta(j') \neq i$, so $(\vec{a} \circ \beta)_{j'} \in E$. Since t witnesses $E \lhd D$, taken at the coordinate i', the value lies in E. The last sentence is this applied to any bijection $\beta : V \to [m]$, which exists because V is finite and nonempty. □

Remark 2.3 (The quantifier form matters)**.** Definition 2.1 constrains the tuple $(z_1, \ldots, z_m)$ rather than quantifying over a separate list $e_1, \ldots, e_m \in E$ of which one entry is then overwritten. For $E \neq \varnothing$ the two readings agree. They differ only when $E = \varnothing$: the overwrite reading is then vacuous for every $m \geq 1$, while the tuple reading is vacuous for $m \geq 2$ and, for $m = 1$, asserts $t^{\mathbf{A}}(z_1) \in \varnothing$ for all $z_1 \in D$ — which is vacuously true if $D = \varnothing$ and false otherwise. So the readings part company exactly at $E = \varnothing$, $m = 1$, $D \neq \varnothing$.

The tuple reading is the correct one, in the sense that it is the one for which Theorem 3.11 is true. Take $S = \varnothing$, $m = 1$ and $D = A$: the relation $R = A$ is S-essential of arity 1, being nonempty by Convention 1.2(b) and disjoint from S, so the theorem must deny 1-ary absorption. This is the one place where the nonemptiness of A is doing work in this remark.

Remark 2.4 (Empty and full)**.** $A \lhd \mathbf{A}$ holds, witnessed by x_1, and $\varnothing \lhd \mathbf{A}$ holds, witnessed by any term of arity at least 2. Convention 1.5 is what keeps the empty subuniverse from trivializing the hypothesis "$\mathbf{B}$ has no proper binary absorbing subalgebra".

Definition 2.5 (Taylor identities)**.** Let $\mathbf{A}$ be an algebra, $D \subseteq A$, $t \in T_\sigma(k)$ and $i \in [k]$. A *Taylor identity for t at i over D* is a pair of maps $u, v : [k] \to \{1, 2\}$ with $u(i) = 1$ and $v(i) = 2$ such that

$$t^{\mathbf{A}}\big(w_{u(1)}, \ldots, w_{u(k)}\big) = t^{\mathbf{A}}\big(w_{v(1)}, \ldots, w_{v(k)}\big) \qquad \text{for all } x, y \in D,$$

where $w_1 = x$ and $w_2 = y$. We say t is *Taylor on D* if a Taylor identity for t at i over D exists for every $i \in [k]$.

Remark 2.6 (Relation to the usual definition)**.** An algebra is usually called *Taylor* if it has an idempotent term satisfying, for each coordinate, an identity in two variables whose two sides carry x and y respectively in that coordinate and arbitrary members of $\{x, y\}$ elsewhere. Definition 2.5 is that condition, relativized to a subset D and stated for a term of the ambient signature; this is the form in which it is used, since in Theorem 5.1 the term is applied to elements of A while its identities are needed only on B.

The next lemma is the engine that converts a one-sided closure condition into two-sided binary absorption. It is the only place a Taylor identity is used.

Lemma 2.7 (One-sided closure gives binary absorption)**.** *Let $\mathbf{A}$ be an algebra, let $E \subseteq D \subseteq A$ with $D \leq \mathbf{A}$ and $E \leq \mathbf{A}$, let $t \in T_\sigma(k)$ and $i \in [k]$, and suppose:*

(a) *there is a Taylor identity (u, v) for t at i over D, and*
(b) *$t^{\mathbf{A}}(z_1, \ldots, z_k) \in E$ whenever $z_i \in E$ and $z_j \in D$ for all $j \neq i$.*

Then $E \lhd_{\mathrm{bin}} D$ in the sense of Definition 2.1; explicitly, the binary term $f := t[u] \in T_\sigma(2)$, the substitution of Definition 1.10 along the map $u : [k] \to [2]$, satisfies $f^{\mathbf{A}}(E \times D) \cup f^{\mathbf{A}}(D \times E) \subseteq E$.

Proof. Let $x, y \in D$ and put $w_1 = x$, $w_2 = y$. By Lemma 1.11 and hypothesis (a),

$$f^{\mathbf{A}}(x, y) = t^{\mathbf{A}}\big(w_{u(1)}, \ldots, w_{u(k)}\big) = t^{\mathbf{A}}\big(w_{v(1)}, \ldots, w_{v(k)}\big).$$

Take $e \in E$ and $d \in D$. For $f^{\mathbf{A}}(e,d)$ set $x = e$, $y = d$ and use the u-form: its ith argument is $w_{u(i)} = w_1 = e \in E$, and every argument is one of e, d, hence lies in D since $E \subseteq D$. By (b), $f^{\mathbf{A}}(e,d) \in E$. For $f^{\mathbf{A}}(d,e)$ set $x = d$, $y = e$ and use the v-form: its ith argument is $w_{v(i)} = w_2 = e \in E$, and again every argument lies in D. By (b), $f^{\mathbf{A}}(d,e) \in E$. $\square$

Definition 2.8 (Star powers)**.** Let $t \in T_\sigma(k)$ with $k \geq 1$. For $\ell \geq 0$ put

$$V_\ell := [k]^{[\ell]} = \{\, \text{functions } [\ell] \to [k] \,\},$$

a finite nonempty set with $|V_\ell| = k^\ell$; V_0 is the one-element set containing the empty function. For $j \in [k]$ let $\rho^\ell_j : V_\ell \to V_{\ell+1}$ be the *block inclusion* that prepends j:

$$\rho^\ell_j(q) := j \frown q, \qquad (j \frown q)(1) = j, \quad (j \frown q)(r+1) = q(r) \ \text{ for } r \in [\ell].$$

Define $t^{*0} := x_\varnothing \in T_\sigma(V_0)$, the unique variable, and, for $\ell \geq 0$,

$$t^{*(\ell+1)} := t\big[\, t^{*\ell}[\rho^\ell_1],\ \ldots,\ t^{*\ell}[\rho^\ell_k] \,\big] \ \in\ T_\sigma(V_{\ell+1}),$$

a substitution in the sense of Definition 1.10. Every $p \in V_{\ell+1}$ is uniquely $p = j \frown q$, with $j := p(1) \in [k]$ and $q \in V_\ell$ given by $q(r) := p(r+1)$ for $r \in [\ell]$; we call j the *block* of p and q its *position in the block*. Each ρ^ℓ_j is injective, and ρ^ℓ_j and $\rho^\ell_{j'}$ have disjoint images for $j \neq j'$, since the block can be read back off as $p(1)$. By Lemma 1.11, for $\vec{a} \in A^{V_{\ell+1}}$,

$$\big(t^{*(\ell+1)}\big)^{\mathbf{A}}(\vec{a}) = t^{\mathbf{A}}(d_1, \ldots, d_k), \qquad d_j = \big(t^{*\ell}\big)^{\mathbf{A}}(\vec{a} \circ \rho^\ell_j).$$

Remark 2.9 (Why the variables are functions)**.** Indexing the variables of $t^{*\ell}$ by $[k^\ell]$ instead would make the block decomposition an instance of Euclidean division: $p = (j-1)k^\ell + q$ with $j \in [k]$, $q \in [k^\ell]$, together with the uniqueness of that decomposition. Indexing them by $V_\ell = [k]^{[\ell]}$ replaces division by reading off the first argument of a function, and the uniqueness of the decomposition becomes the statement that a function on $[\ell+1]$ is determined by its value at 1 and its restriction to the rest. Nothing downstream needs the numeric arity k^ℓ: Theorem 5.2 concludes that $t^{*\ell}$ witnesses $C \lhd \mathbf{A}$ in the sense of Definition 2.1, and Lemma 2.2 converts that to a witness of numeric arity wherever one is wanted. This costs the manuscript one lemma and saves it an imported divisibility fact.

Part II
Essential relations and the arity of absorption

3 Essential relations

Definition 3.1 (Essential relation)**.** Let $\mathbf{A}_1, \ldots, \mathbf{A}_m$ be algebras with $m \geq 1$ and let $S_i \leq \mathbf{A}_i$. A relation $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ is $(S_1, \ldots, S_m)$*-essential* if

$$R \cap (S_1 \times \cdots \times S_{i-1} \times A_i \times S_{i+1} \times \cdots \times S_m) \neq \varnothing \quad \text{for every } i \in [m], \tag{3.1}$$

$$R \cap (S_1 \times \cdots \times S_m) = \varnothing. \tag{3.2}$$

When all $\mathbf{A}_i$ equal $\mathbf{A}$ and all S_i equal S, we say R is S*-essential of arity* m.

Definition 3.2 (Essential for a partition)**.** Let $S \leq \mathbf{A}$, let I be a finite set and let $\mathcal{P} = (I_1, \dots, I_m)$ be a partition of I into $m \geq 1$ nonempty blocks. Writing $S^I := \{x \in A^I : x_i \in S \text{ for all } i \in I\}$, a relation $R \leq \mathbf{A}^I$ is S-*essential for* $\mathcal{P}$ if

$$R \cap \{x \in A^I : x_i \in S \text{ for all } i \in I \setminus I_j\} \neq \varnothing \quad \text{for every } j \in [m], \tag{3.3}$$

$$R \cap S^I = \varnothing. \tag{3.4}$$

If $I = [m]$ and $\mathcal{P}$ is the partition into the singletons $\{1\}, \dots, \{m\}$ in this order, this is exactly S-essentiality of arity m in the sense of Definition 3.1.

Remark 3.3 (The blocks need not be assumed nonempty)**.** Nonemptiness of the blocks is a consequence of (3.3) and (3.4), not an extra hypothesis. If $I_j = \varnothing$ then $\{x \in A^I : x_i \in S \text{ for all } i \in I \setminus I_j\}$ is exactly S^I, so (3.3) at j asserts $R \cap S^I \neq \varnothing$, which (3.4) denies. The same applies to Lemma 3.8, which inherits the hypothesis from here.

Both formalizations derive it rather than assuming it, and the saving is not local: the proof of Theorem 3.11 checks that the blocks $X_1, \dots, X_m$ it builds are nonempty, and that check is the only reason the three degenerate cases $S = A$, $S = \varnothing$ with $m \geq 2$, and $S = \varnothing$ with $m = 1$ are separated off at the start of that proof. With nonemptiness derived, the case analysis disappears.

Lemma 3.4 (Essentiality forces nonemptiness)**.** *If $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ is $(S_1, \dots, S_m)$-essential and $m \geq 2$, then $S_i \neq \varnothing$ for every $i \in [m]$.*

Proof. Fix i and choose $i' \neq i$ in $[m]$. The witness supplied by (3.1) for the index i' is a tuple whose ith coordinate lies in S_i. ☐

Proposition 3.5 (Passing to smaller arity)**.** *Let $R \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_m$ be $(S_1, \dots, S_m)$-essential with $m \geq 2$. Then*

$$R^\flat := \pi_{[m-1]}(R \cap (A_1 \times \cdots \times A_{m-1} \times S_m))$$

is an $(S_1, \dots, S_{m-1})$-essential relation $R^\flat \leq \mathbf{A}_1 \times \cdots \times \mathbf{A}_{m-1}$. Consequently, if an S-essential relation of arity m exists, then S-essential relations of every arity m' with $1 \leq m' \leq m$ exist.

Proof. $R^\flat$ is a subuniverse by Lemma 1.19(b),(c).

(3.1): fix $i \leq m-1$ and let $\vec{x}$ be the witness for index i supplied by (3.1) for R. Its last coordinate lies in S_m, so $\vec{x} \in R \cap (A_1 \times \cdots \times A_{m-1} \times S_m)$, and $\pi_{[m-1]}(\vec{x})$ lies in the required box.

(3.2): suppose $\vec{y} \in R^\flat \cap (S_1 \times \cdots \times S_{m-1})$. Then $\vec{y} = \pi_{[m-1]}(\vec{x})$ for some $\vec{x} \in R$ with $x_m \in S_m$, whence $\vec{x} \in R \cap (S_1 \times \cdots \times S_m)$, contradicting (3.2) for R.

The final sentence follows by iterating, each step reducing the arity by one. ☐

Proposition 3.6 (Absorption forbids essential relations)**.** *Let $S \leq \mathbf{A}$ and suppose $t \in T_\sigma(m)$ witnesses $S \lhd \mathbf{A}$. Then there is no S-essential relation of arity m.*

Proof. Suppose $R \leq \mathbf{A}^m$ is S-essential. For each $i \in [m]$ let $\vec{r}^{(i)} \in R$ be the witness supplied by (3.1), so $r_i^{(i)} \in A$ and $r_j^{(i)} \in S$ for $j \neq i$. Apply $t^{\mathbf{A}^m}$ to the m tuples $\vec{r}^{(1)}, \dots, \vec{r}^{(m)}$; by Lemma 1.14 the result $\vec{z}$ lies in R, and its jth coordinate is

$$z_j = t^{\mathbf{A}}(r_j^{(1)}, \dots, r_j^{(m)}) = t^{\mathbf{A}}(\underbrace{r_j^{(1)}}_{\in S}, \dots, \underbrace{r_j^{(j)}}_{\in A}, \dots, \underbrace{r_j^{(m)}}_{\in S}) \in S$$

by Definition 2.1, the free argument sitting in position j. Hence $\vec{z} \in R \cap S^m$, contradicting (3.2). ☐

Corollary 3.7 (Absorption bounds essential arity)**.** *If $S \lhd \mathbf{A}$ is witnessed by a term of arity m, then every S-essential relation has arity at most $m-1$. In particular the set of arities of S-essential relations is bounded.*

Proof. An S-essential relation of arity $m' \geq m$ would yield one of arity exactly m by Proposition 3.5, contradicting Proposition 3.6. ☐

The converse of Proposition 3.6 is the substantial result of this part. It goes through a grouped form of essentiality.

Lemma 3.8 (Regrouping)**.** *Let $S \leq \mathbf{A}$, let I be a finite set and let $\mathcal{P} = (I_1, \dots, I_m)$ be a partition of I into $m \geq 1$ nonempty blocks. If $R \leq \mathbf{A}^I$ is S-essential for $\mathcal{P}$ in the sense of Definition 3.2, then there exists an S-essential relation of arity m.*

Proof. By strong induction on $|I|$, with m held fixed. Each recursive call below is to a partition of a strictly smaller finite set into the same number m of nonempty blocks; these are the only facts about the measure that are used.

Base case: every block is a singleton. Let $e : [m] \to I$ send j to the unique element of I_j; this is a bijection because the I_j partition I. Reindexing along e (Lemma 1.19(e)) carries R to a subuniverse $R' \leq \mathbf{A}^m$, the box of (3.3) for block j to $S^{j-1} \times A \times S^{m-j}$, and S^I to S^m. So R' is S-essential of arity m.

Inductive step. Otherwise some block has more than one element. Fix j with $|I_j| > 1$ and pick $u \in I_j$; no reordering of the blocks is required. Put $J := (I \setminus I_j) \cup \{u\}$ and consider

$$R_1 := \pi_{I\setminus\{u\}}(R \cap \{x \in A^I : x_u \in S\}) \;\leq\; \mathbf{A}^{I\setminus\{u\}},$$

a subuniverse by Lemma 1.19(b),(c), together with the partition $\mathcal{P}_1$ of $I \setminus \{u\}$ whose jth block is $I_j \setminus \{u\}$ — nonempty because $|I_j| > 1$ — and whose other blocks are those of $\mathcal{P}$.

Condition (3.4) *for R_1.* If $y \in R_1$ has every coordinate in S, then $y = \pi_{I\setminus\{u\}}(x)$ for some $x \in R$ with $x_u \in S$, so $x \in R \cap S^I$, contradicting (3.4) for R.

Condition (3.3) *for R_1 at a block $l \neq j$.* The witness $x \in R$ for I_l has $x_i \in S$ for every $i \in I \setminus I_l$; since $u \in I_j$ and $j \neq l$, in particular $x_u \in S$. Hence $\pi_{I\setminus\{u\}}(x)$ witnesses (3.3) for the lth block of $\mathcal{P}_1$.

Condition (3.3) *for R_1 at block j.* This asks for $x \in R$ with $x_i \in S$ for every $i \in J$, that is, that

$$\pi_J(R) \cap S^J \neq \varnothing. \tag{†}$$

If (†) holds, then R_1 is S-essential for $\mathcal{P}_1$ and $|I \setminus \{u\}| < |I|$, so the inductive hypothesis applies.

If (†) fails, put $R_2 := \pi_J(R) \leq \mathbf{A}^J$, a subuniverse by Lemma 1.19(c), with the partition $\mathcal{P}_2$ of J whose jth block is $\{u\}$ and whose lth block is I_l for $l \neq j$; these are nonempty and partition J. Condition (3.4) for R_2 is exactly the failure of (†). For (3.3) at a block $l \neq j$, project the witness of R for I_l, noting $J \setminus I_l \subseteq I \setminus I_l$. For (3.3) at the block $\{u\}$, project the witness of R for I_j, noting $J \setminus \{u\} = I \setminus I_j$. Finally $|J| = |I| - |I_j| + 1 < |I|$ because $|I_j| > 1$, so the inductive hypothesis applies. ☐

Remark 3.9 (A weaker regrouping than the source's)**.** Lemma 3.8.6 of [3] concludes more: the m-ary S-essential relation can be taken of the form $\pi_I(R \cap \prod_i D_i)$, where I picks exactly one coordinate out of each block and each D_i is A or S. Only the existence assertion is used downstream, in Theorem 3.11, so only existence is proved here. A formalization that wants the sharper statement — for instance to bound which relational operations must be available — should note that the proof above already produces relations of that shape, since both recursive constructions R_1 and R_2 are of the form "intersect with a box, then project"; making the claim precise costs an extra invariant threaded through the induction, and Lemma 1.19 already records that these two operations suffice.

Lemma 3.10 (The algebra of m-ary term operations)**.** *Let* $\mathbf{A}$ *be finite, so that* A^m *is a finite index set. For every* $m \geq 1$, $\mathrm{Clo}_m(\mathbf{A})$ *is a subuniverse of* $\mathbf{A}^{A^m}$ *containing the* m *coordinate projections* $p_1, \ldots, p_m : A^m \to A$*.*

Proof. $p_i = x_i^{\mathbf{A}} \in \mathrm{Clo}_m(\mathbf{A})$. For $f \in \sigma$ of arity k and $t_1, \ldots, t_k \in T_\sigma(m)$, the pointwise application $f^{\mathbf{A}^{A^m}}(t_1^{\mathbf{A}}, \ldots, t_k^{\mathbf{A}})$ sends $\vec{a} \mapsto f^{\mathbf{A}}(t_1^{\mathbf{A}}(\vec{a}), \ldots, t_k^{\mathbf{A}}(\vec{a})) = \big(f(t_1, \ldots, t_k)\big)^{\mathbf{A}}(\vec{a})$, so it lies in $\mathrm{Clo}_m(\mathbf{A})$. □

Theorem 3.11 (Relational description of absorption, Barto–Kazda [2])**.** *Let* $\mathbf{A}$ *be a finite algebra,* $S \leq \mathbf{A}$*, and* $m \geq 1$*. Then* $S \lhd \mathbf{A}$ *is witnessed by some term of arity* m *if and only if there is no* S*-essential relation of arity* m*.*

Proof. The forward implication is Proposition 3.6.

For the converse, assume there is no S-essential relation of arity m.

If $S = A$, every $t \in T_\sigma(m)$ witnesses $S \lhd \mathbf{A}$, since all its values lie in $A = S$; and $x_1 \in T_\sigma(m)$, so such a t exists.

If $S = \varnothing$ and $m \geq 2$, take $t = x_1 \in T_\sigma(m)$: the witnessing condition of Definition 2.1 is vacuous for it, because the condition requires $z_j \in S = \varnothing$ for the $m-1 \geq 1$ indices $j \neq i$. If $S = \varnothing$ and $m = 1$, the hypothesis is unsatisfiable: $R = A$ is an S-essential relation of arity 1, being nonempty and disjoint from S. So in either case there is nothing to prove.

Assume from now on that $\varnothing \neq S \subsetneq A$.

For $i \in [m]$ set

$$X_i := \{\vec{x} \in A^m \ : \ x_i \in A \setminus S \text{ and } x_j \in S \text{ for } j \neq i\}.$$

Each X_i is nonempty, since S and $A \setminus S$ are nonempty. The X_i are pairwise disjoint, because a tuple in $X_i \cap X_{i'}$ with $i \neq i'$ would have $x_i \in S$ and $x_i \notin S$. Put $X := X_1 \cup \cdots \cup X_m$, a finite set because A is finite, and

$$\widetilde{R} := \pi_X\big(\mathrm{Clo}_m(\mathbf{A})\big) \ \leq \ \mathbf{A}^X,$$

using Lemma 3.10 and Lemma 1.19(c). Concretely, $\widetilde{R}$ is the set of restrictions $t^{\mathbf{A}} \restriction_X$ for $t \in T_\sigma(m)$.

The blocks are populated. Fix $i \in [m]$ and take $t = x_i$, so $t^{\mathbf{A}} = p_i$. For $\vec{x} \in X_j$ with $j \neq i$ we have $x_i \in S$ by the definition of X_j, so $p_i(\vec{x}) \in S$. Hence $p_i \restriction_X$ lies in $\{y \in A^X : y_{\vec{x}} \in S \text{ for all } \vec{x} \in X \setminus X_i\}$, which is condition (3.3) for $\widetilde{R}$ at block i.

Applying regrouping. The sets $X_1, \ldots, X_m$ partition X into nonempty blocks, and the previous paragraph is exactly condition (3.3) for $\widetilde{R}$ and that partition. If (3.4) held as well, $\widetilde{R}$ would be S-essential for the partition, and Lemma 3.8 would yield an S-essential relation of arity m, contrary to hypothesis. Hence $\widetilde{R} \cap S^X \neq \varnothing$: some $t \in T_\sigma(m)$ satisfies $t^{\mathbf{A}}(\vec{x}) \in S$ for every $\vec{x} \in X$.

t *witnesses* $S \lhd \mathbf{A}$*.* Let $i \in [m]$ and let $\vec{x} \in A^m$ satisfy $x_j \in S$ for every $j \neq i$. If $x_i \notin S$ then $\vec{x} \in X_i \subseteq X$, so $t^{\mathbf{A}}(\vec{x}) \in S$ as just shown. If $x_i \in S$ then $\vec{x} \in S^m$ and $t^{\mathbf{A}}(\vec{x}) \in S$ by Lemma 1.14. This is precisely the condition of Definition 2.1. □

Corollary 3.12 (Unbounded arity defeats absorption)**.** *Let* $\mathbf{A}$ *be a finite algebra and* $S \leq \mathbf{A}$ *with* $S \lhd \mathbf{A}$*. Then there is a largest arity of an* S*-essential relation, or no* S*-essential relation at all.*

Proof. Immediate from Corollary 3.7: the arities form a set of positive integers bounded above, hence finite, hence with a largest element if nonempty. □

Part III
Zhuk's centers

4 Left centers and their neighborhoods

Throughout this part $\mathbf{A}$ and $\mathbf{B}$ are finite idempotent algebras with nonempty universes, and $R \le \mathbf{A} \times \mathbf{B}$.

Definition 4.1 (Subdirect)**.** $R \le \mathbf{A} \times \mathbf{B}$ is *subdirect* if $\pi_1(R) = A$ and $\pi_2(R) = B$.

Definition 4.2 (Neighborhood and left center)**.** For $a \in A$ put

$$a + R := \{b \in B \ : \ (a,b) \in R\} \ = \ \pi_2(R \cap (\{a\} \times B)),$$

and define the *left center* of R to be

$$C := \{a \in A \ : \ a + R = B\} = \{a \in A : \{a\} \times B \subseteq R\}.$$

The *right center* of R is the left center of $R^- = \{(b,a) : (a,b) \in R\}$, which is a subuniverse of $\mathbf{B} \times \mathbf{A}$ by the reindexing part of Lemma 1.19(e).

Lemma 4.3 (Basic properties)**.** *Let $R \le \mathbf{A} \times \mathbf{B}$ be subdirect with left center C.*

(a) *$a + R \le \mathbf{B}$ for every $a \in A$, and $a + R \ne \varnothing$.*
(b) *$|a + R| \le |B|$, with equality if and only if $a \in C$.*
(c) *$C \le \mathbf{A}$.*
(d) *For every $t \in T_\sigma(k)$ and $a_1, \dots, a_k \in A$,*

$$t^{\mathbf{A}}(a_1, \dots, a_k) + R \ \supseteq \ \{\, t^{\mathbf{B}}(b_1, \dots, b_k) \ : \ b_j \in a_j + R\,\}.$$

Proof. (a) $\{a\} \le \mathbf{A}$ by Lemma 1.7, so $a + R \le \mathbf{B}$ by Lemma 1.19(b),(c). It is nonempty because $\pi_1(R) = A$.

(b) Immediate from $a + R \subseteq B$ and the definition of C.

(c) Let $f \in \sigma$ have arity k, let $a_1, \dots, a_k \in C$ and let $b \in B$. Then $(a_j, b) \in R$ for every j, so $(f^{\mathbf{A}}(a_1, \dots, a_k), f^{\mathbf{B}}(b, \dots, b)) \in R$; and $f^{\mathbf{B}}(b, \dots, b) = b$ because $\mathbf{B}$ is idempotent. As $b \in B$ was arbitrary, $f^{\mathbf{A}}(a_1, \dots, a_k) \in C$.

(d) If $b_j \in a_j + R$ then $(a_j, b_j) \in R$ for each j, so $(t^{\mathbf{A}}(\vec{a}), t^{\mathbf{B}}(\vec{b})) \in R$ by Lemma 1.14. $\square$

Remark 4.4 (Degenerate centers)**.** The content of what follows is the case $\varnothing \ne C \subsetneq A$, but no proof needs to exclude the degenerate cases, and none does. Precisely:

(a) If $C = A$ then $R = A \times B$ and every assertion is trivial.
(b) If $C = \varnothing$ then the absorption assertions about C are vacuous *provided the witnessing term has arity at least* 2; by Remark 2.3 they are not vacuous at arity 1. This is why Theorem 5.2 is proved in the tuple form of Definition 2.1 rather than by fixing a tuple of elements of C; see Remark 5.3. The prescribed witness $t^{*(|B|-1)}$ has arity $k^{|B|-1}$, which equals 1 exactly when $|B| = 1$ or $k = 1$; and $k = 1$ forces $|B| = 1$, because Lemma 1.13 makes a unary $t^{\mathbf{B}}$ the identity while a Taylor identity at its single coordinate makes it constant on B, and $B \ne \varnothing$. So the arity is 1 only when $|B| = 1$, and there subdirectness gives $C = A \ne \varnothing$: the assertion is then not vacuous but true.
(c) Statements whose conclusion is a non-membership, such as Theorem 6.1, are immediate when $C = \varnothing$ because $\mathrm{Sg}_{\mathbf{A}^2}(\varnothing) = \varnothing$ by Lemma 1.6(d).

5 The enlargement step

Theorem 5.1 (One step towards the center)**.** *Let* $\mathbf{A}, \mathbf{B}$ *be finite idempotent algebras, let* $R \leq \mathbf{A} \times \mathbf{B}$ *be subdirect with left center* C*, and let* $t \in T_\sigma(k)$ *be Taylor on* B*. Assume* $\mathbf{B}$ *has no proper binary absorbing subalgebra. Then for every* $i \in [k]$ *and all* $z_1, \ldots, z_k \in A$ *with* $z_j \in C$ *for every* $j \neq i$*,*

$$|\, t^{\mathbf{A}}(z_1, \ldots, z_k) + R \,| \;\geq\; \min(|B|,\; |z_i + R| + 1).$$

Proof. Write $a := z_i$ and $a^+ := t^{\mathbf{A}}(z_1, \ldots, z_k)$, and set

$$E := \{\, t^{\mathbf{B}}(y_1, \ldots, y_k) \;:\; y_i \in a + R,\; y_j \in B \;(j \neq i) \,\}.$$

Note that no element of C is required to exist: the hypothesis constrains only the coordinates $j \neq i$, and if $C = \varnothing$ and $k \geq 2$ it is unsatisfiable, while if $k = 1$ there are no such coordinates.

Step 1: $a^+ + R \supseteq E$*.* Each z_j with $j \neq i$ lies in C, so $z_j + R = B$, and Lemma 4.3(d) gives exactly this containment.

Step 2: $E \supseteq a + R$*.* For $b \in a + R$ take $y_1 = \cdots = y_k = b$; this is admissible because $a + R \subseteq B$, and $t^{\mathbf{B}}(b, \ldots, b) = b$ by Lemma 1.13.

Step 3: the dichotomy. If $E \neq a + R$ then $|E| \geq |a + R| + 1$ by Step 2, and $|a^+ + R| \geq |E| \geq |a + R| + 1$ by Step 1, which is the claim.

So suppose $E = a + R$. Then $t^{\mathbf{B}}(y_1, \ldots, y_k) \in a + R$ whenever $y_i \in a + R$ and $y_j \in B$ for $j \neq i$. Apply Lemma 2.7 inside $\mathbf{B}$, with $D = B$, with $E_{\text{lem}} = a + R$ — a subuniverse of $\mathbf{B}$ by Lemma 4.3(a) — and with the Taylor identity for t at i over B supplied by Definition 2.5. It yields $a + R \lhd_{\text{bin}} \mathbf{B}$. Now $a + R$ is nonempty by Lemma 4.3(a), so by the hypothesis on $\mathbf{B}$ and Convention 1.5 it is not a proper subuniverse: $a + R = B$. Hence $a = z_i \in C$; since also $z_j \in C$ for $j \neq i$, every coordinate of $(z_1, \ldots, z_k)$ lies in C, and $C \leq \mathbf{A}$ by Lemma 4.3(c), so $a^+ = t^{\mathbf{A}}(z_1, \ldots, z_k) \in C$ by Lemma 1.14. Therefore $|a^+ + R| = |B| \geq \min(|B|, |a + R| + 1)$. □

Theorem 5.2 (Zhuk: the left center absorbs)**.** *Let* $\mathbf{A}, \mathbf{B}$ *be finite idempotent algebras, let* $R \leq \mathbf{A} \times \mathbf{B}$ *be subdirect with left center* C*, and let* $t \in T_\sigma(k)$ *be Taylor on* B*. If* $\mathbf{B}$ *has no proper binary absorbing subalgebra, then the term* $t^{*(|B|-1)}$ *of Definition 2.8 witnesses* $C \lhd \mathbf{A}$*. In particular* $C \lhd \mathbf{A}$*.*

Proof. We prove by induction on $\ell \geq 0$ the following statement, whose quantifier shape is deliberately the one used in Definition 2.1: for every $p \in V_\ell$ and every $\vec{z} \in A^{V_\ell}$ with $z_p \in A$ and $z_r \in C$ for $r \neq p$,

$$|\, (t^{*\ell})^{\mathbf{A}}(\vec{z}) + R \,| \;\geq\; \min(|B|,\; |z_p + R| + \ell). \tag{$*_\ell$}$$

For $\ell = 0$ the set V_0 is a single point, so p is that point, the tuple is the single element $z_p \in A$ subject to no further constraint, and the term is the variable x_p; thus $(*_0)$ reads $|z_p + R| \geq \min(|B|, |z_p + R|)$, which is trivially true.

Assume $(*_\ell)$. Let $p \in V_{\ell+1}$ and let $\vec{z} \in A^{V_{\ell+1}}$ satisfy $z_p \in A$ and $z_r \in C$ for $r \neq p$. Write $p = j \frown q$ with $j = p(1) \in [k]$ and $q \in V_\ell$ as in Definition 2.8, so that

$$(t^{*(\ell+1)})^{\mathbf{A}}(\vec{z}) = t^{\mathbf{A}}(d_1, \ldots, d_k), \qquad d_{j'} = (t^{*\ell})^{\mathbf{A}}(\vec{z} \circ \rho^\ell_{j'}).$$

Block j carries z_p at position q and elements of C elsewhere: indeed $\rho^\ell_j(q) = p$, while $\rho^\ell_j(q') \neq p$ for $q' \neq q$ and $\rho^\ell_{j'}(q') \neq p$ for every $j' \neq j$, the block inclusions being injective with disjoint images (Definition 2.8). So $(*_\ell)$ gives $|d_j + R| \geq \min(|B|, |z_p + R| + \ell)$. Each block $j' \neq j$ carries elements of C only, so $d_{j'} \in C$ by Lemma 4.3(c) and Lemma 1.14. The tuple $(d_1, \ldots, d_k)$ therefore satisfies the hypothesis of Theorem 5.1 at coordinate $i = j$ verbatim

— $d_j \in A$ and $d_{j'} \in C$ for $j' \neq j$, with no element of C demanded at coordinate j itself — and that theorem gives

$$\left|t^{\mathbf{A}}(d_1, \dots, d_k) + R\right| \;\geq\; \min\bigl(|B|,\ |d_j + R| + 1\bigr).$$

Writing $N := |B|$ and $x := |z_p + R| + \ell$, the inductive bound $|d_j + R| \geq \min(N, x)$ gives

$$\min\bigl(N,\ |d_j + R| + 1\bigr) \;\geq\; \min\bigl(N,\ \min(N, x) + 1\bigr) \;=\; \min(N,\ x + 1),$$

the first step by monotonicity of min in its second argument and the second by the identity $\min(N, \min(N, x) + 1) = \min(N, x + 1)$, both recorded in Appendix C. This is $(*_{\ell+1})$.

Now take $\ell = |B| - 1$, let $p \in V_\ell$ and let $\vec{z} \in A^{V_\ell}$ satisfy $z_p \in A$ and $z_r \in C$ for $r \neq p$. Since $|z_p + R| \geq 1$ by Lemma 4.3(a), $(*_\ell)$ gives

$$\left|\bigl(t^{*\ell}\bigr)^{\mathbf{A}}(\vec{z}) + R\right| \;\geq\; \min\bigl(|B|,\ 1 + |B| - 1\bigr) = |B|,$$

so the value lies in C by Lemma 4.3(b). This is precisely the condition of Definition 2.1 for the term $t^{*\ell}$, so $t^{*\ell}$ witnesses $C \lhd \mathbf{A}$. ☐

Remark 5.3 (Why the tuple form matters here)**.** Stating $(*_\ell)$ as "for all $\vec{c} \in C^{V_\ell}$, replacing c_p by an arbitrary $a \in A \dots$" would make the induction unusable when $C = \varnothing$: at $\ell = 0$ the hypothesis would quantify over an empty set while the conclusion mentions only a, and the final step would then have to bridge a vacuous statement to the non-vacuous unary instance of Definition 2.1. In the form above no such gap arises. For $C = \varnothing$ and $k \geq 2$ the hypothesis of $(*_\ell)$ is unsatisfiable for $\ell \geq 1$, matching the fact that absorption by a term of arity ≥ 2 is then vacuous; and for $|B| = 1$ we get $\ell = 0$ and the conclusion $A \subseteq C$, which is correct because subdirectness forces $R = A \times B$ and $C = A$. No case analysis is needed.

6 The center is central

Theorem 6.1 (Zhuk: centrality of the left center)**.** *Let $\mathbf{A}, \mathbf{B}$ be finite idempotent algebras and let $R \leq \mathbf{A} \times \mathbf{B}$ be subdirect with left center C. Assume $\mathbf{B}$ has no proper binary absorbing subalgebra. Then for every $a \in A \setminus C$,*

$$(a, a) \;\notin\; \mathrm{Sg}_{\mathbf{A}^2}\Bigl(\bigl(\{a\} \times C\bigr) \cup \bigl(C \times C\bigr) \cup \bigl(C \times \{a\}\bigr)\Bigr).$$

Proof. Write $G := (\{a\} \times C) \cup (C \times C) \cup (C \times \{a\})$ and suppose $(a, a) \in \mathrm{Sg}_{\mathbf{A}^2}(G)$. If $C = \varnothing$ then $G = \varnothing$ and $\mathrm{Sg}_{\mathbf{A}^2}(\varnothing) = \varnothing$ by Lemma 1.6(d); so $C \neq \varnothing$ and $G \neq \varnothing$.

Normal form for the generators. The three sets $\{a\} \times C$, $C \times C$ and $C \times \{a\}$ are pairwise disjoint: a pair lying in two of them would force $a \in C$. Enumerate G so that the elements of $\{a\} \times C$ come first, then those of $C \times C$, then those of $C \times \{a\}$; such an enumeration exists by Lemma 1.20. By Lemma 1.15 there are $n \geq 1$, an enumeration $g_1, \dots, g_n$ of G of this shape, and a term $t \in T_\sigma(n)$ with $(a, a) = t^{\mathbf{A}^2}(g_1, \dots, g_n)$. Let $0 \leq p \leq q \leq n$ be such that

$$g_r = (a, \gamma_r)\ (r \leq p), \qquad g_r = (\gamma'_r, \gamma_r)\ (p < r \leq q), \qquad g_r = (\gamma'_r, a)\ (r > q),$$

with all $\gamma_r, \gamma'_r \in C$. Here $p \leq q$ because the three blocks are listed consecutively in this order; moreover $p \geq 1$ and $q < n$, because $C \neq \varnothing$ makes the first and third blocks nonempty. Only $p \leq q$ is used below. Reading the two coordinates of $(a, a) = t^{\mathbf{A}^2}(g_1, \dots, g_n)$ separately gives

$$a = t^{\mathbf{A}}\bigl(\underbrace{a, \dots, a}_{p},\ \gamma'_{p+1}, \dots, \gamma'_n\bigr), \tag{6.1}$$

$$a = t^{\mathbf{A}}\bigl(\gamma_1, \dots, \gamma_q,\ \underbrace{a, \dots, a}_{n-q}\bigr). \tag{6.2}$$

Claim 1. If $z_1,\dots,z_p \in a+R$ and $z_{p+1},\dots,z_n \in B$, then $t^{\mathbf{B}}(z_1,\dots,z_n) \in a+R$.

Indeed, $(a,z_r) \in R$ for $r \le p$ because $z_r \in a+R$, and $(\gamma'_r, z_r) \in R$ for $r > p$ because $\gamma'_r \in C$ and $z_r \in B$. Applying t coordinatewise (Lemma 1.14) and using (6.1) for the first coordinate gives $\big(a,\, t^{\mathbf{B}}(\vec{z})\big) \in R$, i.e. $t^{\mathbf{B}}(\vec{z}) \in a+R$.

Claim 2. If $z_1,\dots,z_q \in B$ and $z_{q+1},\dots,z_n \in a+R$, then $t^{\mathbf{B}}(z_1,\dots,z_n) \in a+R$.

Indeed, $(\gamma_r, z_r) \in R$ for $r \le q$ because $\gamma_r \in C$ and $z_r \in B$, and $(a,z_r) \in R$ for $r > q$. Applying t coordinatewise and using (6.2) gives $\big(a,\, t^{\mathbf{B}}(\vec{z})\big) \in R$.

Conclusion. Choose any m with $p \le m \le q$; such an m exists because $p \le q$. Let $\mu : [n] \to [2]$ send $r \mapsto 1$ for $r \le m$ and $r \mapsto 2$ for $r > m$, and put $f := t[\mu] \in T_\sigma(2)$, so that by Lemma 1.11

$$f^{\mathbf{B}}(y_1,y_2) = t^{\mathbf{B}}(\underbrace{y_1,\dots,y_1}_{m}, \underbrace{y_2,\dots,y_2}_{n-m}).$$

For $e \in a+R$ and $b \in B$: in $f^{\mathbf{B}}(e,b)$ the first m arguments equal e and the rest equal b; since $m \ge p$, the first p arguments lie in $a+R$ and all arguments lie in B, so Claim 1 gives $f^{\mathbf{B}}(e,b) \in a+R$. For $b \in B$ and $e \in a+R$: in $f^{\mathbf{B}}(b,e)$ the first m arguments equal b and the rest equal e; since $m \le q$, the arguments in positions $q+1,\dots,n$ all equal $e \in a+R$, and all arguments lie in B, so Claim 2 gives $f^{\mathbf{B}}(b,e) \in a+R$.

Hence $a+R \lhd_{\mathrm{bin}} \mathbf{B}$. By Lemma 4.3(a) it is nonempty, and it is a proper subuniverse of B because $a \notin C$. This contradicts the hypothesis on $\mathbf{B}$ together with Convention 1.5. ☐

Definition 6.2 (Central absorption)**.** Let $C \le \mathbf{A}$. We say C *centrally absorbs* $\mathbf{A}$, written $C \lhd_Z \mathbf{A}$, if

(a) $C \lhd \mathbf{A}$, and
(b) for every $a \in A \setminus C$, $(a,a) \notin \mathrm{Sg}_{\mathbf{A}^2}\big((\{a\}\times C) \cup (C \times \{a\})\big)$.

Corollary 6.3 (Zhuk's center theorem)**.** *Let $\mathbf{A},\mathbf{B}$ be finite idempotent algebras and $R \le \mathbf{A}\times\mathbf{B}$ subdirect with left center C. If some $t \in T_\sigma(k)$ is Taylor on B and $\mathbf{B}$ has no proper binary absorbing subalgebra, then $C \lhd_Z \mathbf{A}$.*

Proof. Condition (a) of Definition 6.2 is Theorem 5.2. For (b), note $(\{a\}\times C)\cup(C\times\{a\}) \subseteq (\{a\}\times C)\cup(C\times C)\cup(C\times\{a\})$, so the generated subuniverses are nested by Lemma 1.6(c), and Theorem 6.1 applies. ☐

7 The essential doubling trick

Lemma 7.1 (Zhuk–Kozik essential doubling)**.** *Let $n \ge 1$ and let $\mathbf{A}_0,\dots,\mathbf{A}_{n+1}$ be algebras with $\mathbf{A}_{n+1}$ finite and idempotent. Let $C \le \mathbf{A}_0$, $B_i \le \mathbf{A}_i$ for $i \in [n]$, and $C' \le \mathbf{A}_{n+1}$ with $C' \lhd_Z \mathbf{A}_{n+1}$. If there exists a $(C,B_1,\dots,B_n,C')$-essential relation $R \le \mathbf{A}_0\times\cdots\times\mathbf{A}_{n+1}$, then there exists a $(C,B_1,\dots,B_n,B_n,\dots,B_1,C)$-essential relation*

$$R' \;\le\; \mathbf{A}_0\times\cdots\times\mathbf{A}_n\times\mathbf{A}_n\times\cdots\times\mathbf{A}_0$$

of arity $2n+2$.

Proof. Throughout the proof write

$$D_0 := C, \qquad D_r := B_r \ (1 \le r \le n), \qquad D_{n+1} := C',$$

so that the designated subuniverses of the hypothesis are $D_0,\dots,D_{n+1}$ and the box of (3.1) at index i is $\prod_{r\ne i} D_r$ with A_i inserted at position i. This avoids the schematic ellipsis $C\times\cdots\times A_i\times\cdots\times C'$, which is ambiguous at $i=0$ and $i=n+1$, where the free coordinate replaces C or C' rather than some B_r.

For a relation $P \leq \mathbf{A}_0 \times \cdots \times \mathbf{A}_{n+1}$ write

$$\beta(P) := \pi_{n+1}\big(P \cap (D_0 \times \cdots \times D_n \times A_{n+1})\big) \ \leq \ \mathbf{A}_{n+1},$$

a subuniverse by Lemma 1.19(b),(c).

Step 0: choose R minimizing $|\beta(R)|$. The family of $(C, B_1, \ldots, B_n, C')$-essential relations is nonempty by hypothesis, and $|\beta(P)|$ is a natural number bounded by $|A_{n+1}|$; so the set of values $\{|\beta(P)| : P \text{ is } (C, B_1, \ldots, B_n, C')\text{-essential}\}$ is a nonempty set of naturals bounded by $|A_{n+1}|$, and therefore has a least element. Some essential P attains that least value, because the value belongs to the displayed set; name one such P as R and write $B' := \beta(R)$. Only the finiteness of $\mathbf{A}_{n+1}$ is used: the family of essential relations need not be finite, and no choice principle is involved, since a single attaining relation is named by existential instantiation.

By (3.1) at index $n+1$, $B' \neq \varnothing$. By (3.2), $B' \cap C' = \varnothing$: an element of the intersection would be the last coordinate of a tuple of R lying in $D_0 \times \cdots \times D_n \times D_{n+1}$. By Lemma 3.4, C, the B_i and C' are all nonempty.

Step 1: the generation property of B'. We prove

$$\text{for all } b_1, b_2 \in B' : \qquad b_2 \in \mathrm{Sg}_{\mathbf{A}_{n+1}}(C' \cup \{b_1\}). \tag{‡}$$

Let $b_1 \in B'$ be arbitrary. For each $i \in \{0, \ldots, n\}$ let $\vec{x}^{\,(i)} \in R$ be the witness of (3.1) at index i, so $x_r^{(i)} \in D_r$ for every $r \neq i$; in particular $x_{n+1}^{(i)} \in D_{n+1} = C'$ because $i \leq n$. Let $\vec{y} \in R \cap (D_0 \times \cdots \times D_n \times A_{n+1})$ have $y_{n+1} = b_1$, which exists because $b_1 \in B'$. Put

$$R'' := \mathrm{Sg}_{\mathbf{A}_0 \times \cdots \times \mathbf{A}_{n+1}}(\{\vec{x}^{\,(0)}, \ldots, \vec{x}^{\,(n)}, \vec{y}\}) \subseteq R.$$

Then R'' is $(C, B_1, \ldots, B_n, C')$-essential: (3.1) at index $i \leq n$ is witnessed by $\vec{x}^{\,(i)}$, (3.1) at index $n+1$ by $\vec{y}$, and (3.2) holds because $R'' \subseteq R$. So R'' is a legitimate competitor in the very minimization of Step 0 — the designated subuniverses $C, B_1, \ldots, B_n, C'$ are the same ones, and only the relation varies. By minimality $|\beta(R'')| \geq |B'|$, while $\beta(R'') \subseteq \beta(R) = B'$ because $R'' \subseteq R$; hence $\beta(R'') = B'$. Combining this with Lemma 1.18(c) applied to π_{n+1}, and with Lemma 1.6(c) for the last inclusion, gives the chain

$$\begin{aligned} B' \ = \ \beta(R'') \ &\subseteq \ \pi_{n+1}(R'') \\ &= \ \mathrm{Sg}_{\mathbf{A}_{n+1}}(\{x_{n+1}^{(0)}, \ldots, x_{n+1}^{(n)}, b_1\}) \\ &\subseteq \ \mathrm{Sg}_{\mathbf{A}_{n+1}}(C' \cup \{b_1\}), \end{aligned}$$

the last step because every $x_{n+1}^{(i)}$ lies in C'. As $b_1 \in B'$ was arbitrary and every $b_2 \in B'$ lies in the left-hand side, this is (‡).

Step 2: the linking relation. Fix from now on one particular element $b \in B'$, which exists because $B' \neq \varnothing$, and set

$$S := \mathrm{Sg}_{\mathbf{A}_{n+1}^2}\big((\{b\} \times C') \cup (C' \times \{b\})\big) \ \leq \ \mathbf{A}_{n+1}^2.$$

The source describes S as a symmetric relation. That plays no role below — the argument draws on the two generating blocks $\{b\} \times C'$ and $C' \times \{b\}$ separately — so symmetry is neither asserted nor proved here, and a formalization need not establish it.

Step 3: $S \cap (B' \times B') = \varnothing$. Suppose $(b', b'') \in S$ with $b', b'' \in B'$.

Let $P := \pi_1(S \cap (A_{n+1} \times B')) \leq \mathbf{A}_{n+1}$, the set of elements having an S-neighbor in B', a subuniverse by Lemma 1.19(b),(c). Every $c \in C'$ lies in P, since $(c, b) \in S$ and $b \in B'$; and $b' \in P$, since $(b', b'') \in S$ and $b'' \in B'$. Hence $\mathrm{Sg}_{\mathbf{A}_{n+1}}(C' \cup \{b'\}) \subseteq P$ by Lemma 1.6(c). Instantiating (‡) at $b_1 := b'$ and $b_2 := b$ — legitimate since $b', b \in B'$ — gives $b \in \mathrm{Sg}_{\mathbf{A}_{n+1}}(C' \cup \{b'\})$, so $b \in P$: there is some $b''' \in B'$ with $(b, b''') \in S$.

Let $Q := \pi_2(S \cap (\{b\} \times A_{n+1})) \leq \mathbf{A}_{n+1}$, using $\{b\} \leq \mathbf{A}_{n+1}$ from Lemma 1.7. Every $c \in C'$ lies in Q, since $(b, c) \in S$; and $b''' \in Q$. Hence $\mathrm{Sg}_{\mathbf{A}_{n+1}}(C' \cup \{b'''\}) \subseteq Q$ by Lemma 1.6(c). Instantiating (‡) a second time, now at $b_1 := b'''$ and $b_2 := b$ — legitimate since $b''', b \in B'$ — gives $b \in \mathrm{Sg}_{\mathbf{A}_{n+1}}(C' \cup \{b'''\})$, so $b \in Q$, that is, $(b, b) \in S$.

But $b \notin C'$, because $b \in B'$ and $B' \cap C' = \varnothing$. So $(b, b) \in S$ contradicts condition (b) of Definition 6.2 for $C' \lhd_Z \mathbf{A}_{n+1}$, whose generating set is exactly the one defining S.

Remark 7.2 (The shape of Step 3)**.** Step 3 is the only argument in this document that doubles back on itself, and it is worth recording the dependency explicitly. (‡) is a *universally quantified* statement about ordered pairs from B', established in Step 1 before any element of Step 3 is named. Step 3 then uses it twice, at the pairs (b', b) and (b''', b), and the second of these applies it to b''', an element that Step 3 itself produced from the first application. There is no circularity: b''' is supplied with the side condition $b''' \in B'$, which is all (‡) requires of its arguments. A formalization should therefore package (‡) as a standalone, universally quantified lemma, proved before the element b of Step 2 is fixed. That is also how [3] states it, immediately before specializing b; the gain here is only that the two later instantiations, including the one at the newly produced b''', are named rather than left to the reader.

Step 4: the doubled relation. Define

$$R' := \{(x_0, \ldots, x_n, y_n, \ldots, y_0) \ : \ \exists\, x_{n+1}, y_{n+1} \in A_{n+1} \ \ \vec{x} \in R,\ \vec{y} \in R,\ (x_{n+1}, y_{n+1}) \in S\},$$

where $\vec{x} = (x_0, \ldots, x_{n+1})$ and $\vec{y} = (y_0, \ldots, y_{n+1})$. It is a subuniverse of $\mathbf{A}_0 \times \cdots \times \mathbf{A}_n \times \mathbf{A}_n \times \cdots \times \mathbf{A}_0$: form the relation

$$T := \{(\vec{x}, \vec{y}) \in \textstyle\prod_{i=0}^{n+1} A_i \times \prod_{i=0}^{n+1} A_i \ : \ \vec{x} \in R,\ \vec{y} \in R,\ (x_{n+1}, y_{n+1}) \in S\},$$

which is an intersection of three cylinders — the preimages of R, of R and of S under the projections onto the $\vec{x}$ coordinates, the $\vec{y}$ coordinates, and the pair (x_{n+1}, y_{n+1}) respectively — each a subuniverse of the $(2n+4)$-fold product by Lemma 1.19(d), so T is a subuniverse by Lemma 1.19(a). Now project away the two coordinates carrying x_{n+1}, y_{n+1} and reorder the surviving y coordinates into the order $y_n, \ldots, y_0$, using Lemma 1.19(c),(e).

Label the coordinates of R' as $0, \ldots, n$ on the x-side and $n^\dagger, \ldots, 0^\dagger$ on the y-side, with designated subuniverses $D_0, \ldots, D_n$ and $D_n, \ldots, D_0$ respectively.

(3.1) *at an x-side coordinate $i \in \{0, \ldots, n\}$*: take $\vec{x} = \vec{x}^{(i)}$, so $x_{n+1} \in C'$, and take $\vec{y} \in R \cap (D_0 \times \cdots \times D_n \times \{b\})$, which exists because $b \in B'$. Then $(x_{n+1}, b) \in C' \times \{b\} \subseteq S$, so the resulting tuple lies in R'; its x-side has the free coordinate at i and designated entries elsewhere, and its y-side is entirely designated.

(3.1) *at a y-side coordinate*: symmetric, exchanging the roles of $\vec{x}$ and $\vec{y}$ and using $\{b\} \times C' \subseteq S$.

(3.2): suppose $(x_0, \ldots, x_n, y_n, \ldots, y_0) \in R'$ with $x_r, y_r \in D_r$ for every $r \in \{0, \ldots, n\}$, witnessed by x_{n+1}, y_{n+1}. Then $x_{n+1} \in \beta(R) = B'$ and $y_{n+1} \in B'$ by the definition of β, while $(x_{n+1}, y_{n+1}) \in S$; this contradicts Step 3. □

Remark 7.3 (Only the all-equal instance is used)**.** Lemma 7.1 is stated over a mixed product, with designated subuniverses $C, B_1, \ldots, B_n, C'$ that may differ. Its only consumer, Corollary 8.1, applies it with every factor equal to $\mathbf{A}$ and every designated subuniverse equal to C, and both formalizations state and prove only that specialization. The gain is structural rather than cosmetic: in the specialized form every relation is a subuniverse of a *power*, so a product of a family of algebras — needed nowhere else in this document — is never required, and the conclusion is literally S-essentiality of arity $2n + 2$ in the sense of Definition 3.1.

Step 4 also reverses the second half of the doubled index, writing $\mathbf{A}_0 \times \cdots \times \mathbf{A}_n \times \mathbf{A}_n \times \cdots \times \mathbf{A}_0$. Essentiality does not depend on the order of the coordinates, so neither formalization reverses anything; both index the doubled relation by two copies of $\{0,\dots,n\}$ side by side and reindex once at the end, by Lemma 1.19(e).

8 Central absorption is ternary

Corollary 8.1 (Central absorption is witnessed by a ternary term)**.** *Let* $\mathbf{A}$ *be a finite idempotent algebra and* $C \le \mathbf{A}$ *with* $C \lhd_Z \mathbf{A}$*. Then some* $s \in T_\sigma(3)$ *witnesses* $C \lhd \mathbf{A}$*.*

Proof. Suppose not. By Theorem 3.11 there exists a C-essential relation of arity 3. Since $C \lhd \mathbf{A}$ holds by Definition 6.2(a), Corollary 3.12 provides a largest arity M of a C-essential relation, and $M \ge 3$.

Let $R \le \mathbf{A}^M$ be C-essential of arity M and write $M = n+2$ with $n = M-2 \ge 1$. Then R is $(C, B_1, \dots, B_n, C')$-essential for $\mathbf{A}_0 = \dots = \mathbf{A}_{n+1} = \mathbf{A}$, $B_1 = \dots = B_n = C$ and $C' = C$, and $C' \lhd_Z \mathbf{A}$ by hypothesis. Lemma 7.1 produces a $(C, C, \dots, C)$-essential relation of arity $2n+2 = 2M-2$. Since $M \ge 3$ we have $2M-2 \ge M+1 > M$, contradicting the maximality of M. □

Remark 8.2 (The arities produced)**.** Iterating Lemma 7.1 from arity 3 produces C-essential relations of arities $3, 4, 6, 10, 18, \dots$, that is $2+2^k$, since $2(2+2^k)-2 = 2+2^{k+1}$. Corollary 8.1 is arranged so that only the strict increase $2M-2 > M$ for $M \ge 3$ is needed, which removes the induction.

Proof of Theorem 0.1. By Corollary 6.3, $C \lhd_Z \mathbf{A}$. By Corollary 8.1 there is $s \in T_\sigma(3)$ witnessing $C \lhd \mathbf{A}$, which by Definition 2.1 is exactly the displayed containment. □

Remark 8.3 (What is *not* proved here)**.** Section 3.10 of [3] continues past Corollary 8.1: central absorption is shown to be compatible with quotients, transitively closed and compatible with products, hence a "good absorption concept"; a single ternary term is shown to witness $C \lhd_Z \mathbf{B}$ simultaneously for all $C \lhd_Z \mathbf{B}$ with $\mathbf{B} \in HSP(\mathbf{A})$; and a criterion is given for central absorption to imply binary absorption. Those results, and the Absorption Theorem of [1] that they feed, are outside the scope of this document.

A Statement-level citation index

This appendix records, for every labeled numbered statement, the labeled statements cited in its statement and proof. The table is generated mechanically from the citation commands in the source, so it is a syntactic citation list rather than a semantic closure: a fact used without a citation command does not appear here. Facts imported from outside the manuscript are collected instead in the intended imported-background package of Appendix C.

Two cautions on how to read the table. First, it is a *cross-reference index*, not a logical dependency graph: a row records the labels mentioned in a statement and its proof, including mentions that are expository rather than inferential. Second, several rows therefore cite forward: Definition 1.8 and Convention 1.2 point ahead to the results that fix their design, Remarks 1.12, 1.22, 2.3, 2.6, 3.9 and 4.4 each point ahead to material they motivate or qualify, and Theorem 0.1 names the two definitions its statement uses. The only *proof* dependency that runs forward is that of Theorem 0.1, which is stated at the front and proved at the end.

Statement	Direct internal prerequisites
Theorem 0.1 *Main theorem*	Definition 2.5, Definition 2.1, Corollary 6.3 and Corollary 8.1

Continued on next page

Statement	Direct internal prerequisites
Definition 1.1 *Signature and algebra*	Definitions and imported background (Appendix C) only.
Convention 1.2 *Standing hypotheses*	Lemma 1.6, Convention 1.5, Theorem 6.1, Remark 2.3, Theorem 3.11 and Remark 4.4
Definition 1.3 *Idempotence*	Definitions and imported background (Appendix C) only.
Definition 1.4 *Subuniverse*	Definitions and imported background (Appendix C) only.
Convention 1.5 *Subalgebra means nonempty*	Definitions and imported background (Appendix C) only.
Lemma 1.6 *Intersections and generation*	Definitions and imported background (Appendix C) only.
Lemma 1.7 *Singletons*	Definition 1.3
Definition 1.8 *Products*	Theorem 3.11 and Lemma 7.1
Definition 1.9 *Terms*	Definition 2.8
Definition 1.10 *Simultaneous substitution*	Definitions and imported background (Appendix C) only.
Lemma 1.11 *Evaluation of a substitution*	Definition 1.9
Remark 1.12 *Where substitution is used*	Definition 1.10, Lemma 2.7, Definition 2.8, Theorem 6.1 and Lemma 2.2
Lemma 1.13 *Term operations are idempotent*	Definition 1.3
Lemma 1.14 *Preservation*	Definition 1.4
Lemma 1.15 *Generation by a fixed list*	Lemma 1.14 and Lemma 1.6
Remark 1.16 *Why a fixed list*	Remark 1.12
Definition 1.17 *Homomorphism*	Definitions and imported background (Appendix C) only.
Lemma 1.18 *Images and generation*	Lemma 1.15
Lemma 1.19 *Relational constructions*	Lemma 1.6 and Lemma 1.18
Lemma 1.20 *Block-respecting enumeration*	Definitions and imported background (Appendix C) only.
Remark 1.21 *The enumeration is avoidable*	Lemma 1.20, Theorem 6.1 and Lemma 1.15
Remark 1.22 *The only relational constructions used*	Lemma 1.19, Lemma 7.1 and Lemma 3.8

Continued on next page

Statement	Direct internal prerequisites
Definition 2.1 *Absorption*	Lemma 2.2
Lemma 2.2 *Renaming the variable set*	Lemma 1.11
Remark 2.3 *The quantifier form matters*	Definition 2.1, Theorem 3.11 and Convention 1.2
Remark 2.4 *Empty and full*	Convention 1.5
Definition 2.5 *Taylor identities*	Definitions and imported background (Appendix C) only.
Remark 2.6 *Relation to the usual definition*	Definition 2.5 and Theorem 5.1
Lemma 2.7 *One-sided closure gives binary absorption*	Definition 2.1, Definition 1.10 and Lemma 1.11
Definition 2.8 *Star powers*	Definition 1.10 and Lemma 1.11
Remark 2.9 *Why the variables are functions*	Theorem 5.2, Definition 2.1 and Lemma 2.2
Definition 3.1 *Essential relation*	Definitions and imported background (Appendix C) only.
Definition 3.2 *Essential for a partition*	Definition 3.1
Remark 3.3 *The blocks need not be assumed nonempty*	Lemma 3.8 and Theorem 3.11
Lemma 3.4 *Essentiality forces nonemptiness*	Definitions and imported background (Appendix C) only.
Proposition 3.5 *Passing to smaller arity*	Lemma 1.19
Proposition 3.6 *Absorption forbids essential relations*	Lemma 1.14 and Definition 2.1
Corollary 3.7 *Absorption bounds essential arity*	Proposition 3.5 and Proposition 3.6
Lemma 3.8 *Regrouping*	Definition 3.2 and Lemma 1.19
Remark 3.9 *A weaker regrouping than the source's*	Theorem 3.11 and Lemma 1.19
Lemma 3.10 *The algebra of m-ary term operations*	Definitions and imported background (Appendix C) only.

Continued on next page

Statement	Direct internal prerequisites
Theorem 3.11 *Relational description of absorption, Barto–Kazda [2]*	Proposition 3.6, Definition 2.1, Lemma 3.10, Lemma 1.19, Lemma 3.8 and Lemma 1.14
Corollary 3.12 *Unbounded arity defeats absorption*	Corollary 3.7
Definition 4.1 *Subdirect*	Definitions and imported background (Appendix C) only.
Definition 4.2 *Neighborhood and left center*	Lemma 1.19
Lemma 4.3 *Basic properties*	Lemma 1.7, Lemma 1.19 and Lemma 1.14
Remark 4.4 *Degenerate centers*	Remark 2.3, Theorem 5.2, Definition 2.1, Remark 5.3, Lemma 1.13, Theorem 6.1 and Lemma 1.6
Theorem 5.1 *One step towards the center*	Lemma 4.3, Lemma 1.13, Lemma 2.7, Definition 2.5, Convention 1.5 and Lemma 1.14
Theorem 5.2 *Zhuk: the left center absorbs*	Definition 2.8, Definition 2.1, Lemma 4.3, Lemma 1.14 and Theorem 5.1
Remark 5.3 *Why the tuple form matters here*	Definition 2.1
Theorem 6.1 *Zhuk: centrality of the left center*	Lemma 1.6, Lemma 1.20, Lemma 1.15, Lemma 1.14, Lemma 1.11, Lemma 4.3 and Convention 1.5
Definition 6.2 *Central absorption*	Definitions and imported background (Appendix C) only.
Corollary 6.3 *Zhuk's center theorem*	Definition 6.2, Theorem 5.2, Lemma 1.6 and Theorem 6.1
Lemma 7.1 *Zhuk–Kozik essential doubling*	Lemma 1.19, Lemma 3.4, Lemma 1.18, Lemma 1.6, Lemma 1.7 and Definition 6.2
Remark 7.2 *The shape of Step 3*	Definitions and imported background (Appendix C) only.
Remark 7.3 *Only the all-equal instance is used*	Lemma 7.1, Corollary 8.1, Definition 3.1 and Lemma 1.19
Corollary 8.1 *Central absorption is witnessed by a ternary term*	Theorem 3.11, Definition 6.2, Corollary 3.12 and Lemma 7.1
Remark 8.2 *The arities produced*	Lemma 7.1 and Corollary 8.1
Remark 8.3 *What is* not *proved here*	Corollary 8.1

Continued on next page

Statement	Direct internal prerequisites
Narrative outside labeled statements (prose of Parts I–III)	Theorem 3.11, Definition 1.10, Lemma 1.11, Lemma 1.19, Lemma 3.8, Lemma 7.1, Lemma 2.7, Theorem 5.1, Theorem 5.2, Theorem 6.1, Corollary 8.1, Convention 1.5, Definition 6.2, Remark 3.3, Remark 1.21, Remark 7.3, Lemma 1.14, Definition 1.8, Definition 2.8 and Proposition 3.6

B Suggested module order

The citation index above, together with the imported background of Appendix C, approximates but does not determine the import obligations of each module. A coarser topological order, useful for organizing source files, is:

1. signatures, algebras, subuniverses, generation, indexed products, and the singleton lemma;
2. terms, term operations, simultaneous substitution and its evaluation law, preservation, and generation by a fixed list;
3. homomorphisms, the compatibility of images with generation, and the five relational constructions of Lemma 1.19: intersection, box intersection, projection, cylinder and reindexing;
4. absorption, binary absorption, Taylor identities, and Lemma 2.7;
5. essential relations for a product and for a partition, arity reduction, and the block-regrouping Lemma 3.8;
6. $\mathrm{Clo}_m(\mathbf{A})$ as a subuniverse of $\mathbf{A}^{A^m}$, and the relational description of absorption;
7. neighborhoods $a+R$, left centers, the enlargement step, star powers, and Theorem 5.2;
8. central absorption, Theorem 6.1, the doubling trick, and the ternary collapse.

Modules 1–3 are reusable universal-algebra infrastructure with no reference to absorption. Module 6 is the only place where an algebra of functions ($\mathbf{A}^{A^m}$) is formed, and hence the only place where an exponentially large index set appears; nothing is computed there, so the size is immaterial, but a formalization should expect the type $\mathbf{A}^{A^m}$ rather than a list-based encoding of terms to be the convenient object.

C Imported background

The following background package collects, in the form used, the facts imported but not proved in this manuscript. A formal development should obtain these from its ambient libraries.

1. **Finite sets and cardinality.** For finite sets $X \subseteq Y$: $|X| \leq |Y|$, with equality if and only if $X = Y$; and if $X \subsetneq Y$ then $|X| + 1 \leq |Y|$. A nonempty set of natural numbers bounded above has a greatest element; a nonempty set of natural numbers has a least element.

2. **Arithmetic of** $\min$**.** For natural numbers, $\min$ is monotone in its second argument, so $x \leq y$ implies $\min(N, x{+}1) \leq \min(N, y{+}1)$; and $\min(N, \min(N, x){+}1) = \min(N, x{+}1)$. The induction of Theorem 5.2 uses them in that order: monotonicity first, to push the inductive bound on $|d_j + R|$ through $\min(N, \cdot + 1)$, and the identity second, to collapse the resulting nested $\min$.

3. **Induction.** Structural induction on an inductively defined type of terms; strong induction on the natural numbers.

4. **Products and functions.** Indexed products of sets, tuples and their coordinate projections, restriction of a function to a subset, composition, images and preimages, and the elementary facts that images preserve unions and that a projection of an intersection is contained in the intersection of the projections.

5. **Finite enumeration and choice.** A finite set admits an enumeration $s_1, \ldots, s_N$; and a finite nonempty family of witnesses admits a choice of one witness from each member, as used when selecting the tuples $\vec{x}^{(i)}$ in Lemma 7.1. The block-respecting refinement of the first is not imported: it is proved as Lemma 1.20.

6. **Elementary inequalities on indices.** For the block bookkeeping of Lemma 3.8: if I_j is a block with $|I_j| > 1$ and $u \in I_j$, then $|I \setminus \{u\}| < |I|$ and $|(I \setminus I_j) \cup \{u\}| = |I| - |I_j| + 1 < |I|$.

Simultaneous substitution of terms into terms is deliberately *not* in this list. It is part of the development (Definition 1.10, Lemma 1.11), not of the imported background.

Euclidean division was on this list in earlier drafts, as the indexing underlying Definition 2.8: every $p \in [k^{\ell+1}]$ is uniquely $(j-1)k^\ell + q$. It is no longer imported, because the star powers are now indexed by $V_\ell = [k]^{[\ell]}$ and the block decomposition is the splitting of a function into its first value and the rest; see Remark 2.9. Item 1 above absorbs what remains: the only cardinality fact the change introduces is that a finite nonempty set admits a bijection with some $[m]$, used in Lemma 2.2.

The list is intended to cover the imported inferences, without a claim of demonstrated exhaustiveness; reviewers are invited to report any use of background not covered by an item above.

D Concordance with the source

Section 3.10 of [3] states its results under the numbers in the left column. The correspondence is close but not exact, and the third column records where the statement proved here differs in strength or in hypotheses from the one in the source.

[3]	**This document**	**Difference**
Prop. 3.8.2	Proposition 3.5	Stated for possibly distinct algebras $\mathbf{A}_1, \ldots, \mathbf{A}_m$ and subuniverses S_i; the source states the one-algebra case.
Prop. 3.8.3	Proposition 3.6	None.
Thm. 3.8.5	Theorem 3.11	The source assumes $\mathbf{A}$ finite and idempotent; idempotence is not used here and is dropped. The degenerate cases $S = \varnothing$ and $S = A$ are treated explicitly.

Continued on next page

[3]	This document	Difference
Lem. 3.8.6	Lemma 3.8	*Weaker.* The source also records the specific form $R' = \pi_I(R \cap \prod_i D_i)$ of the resulting relation, with one surviving coordinate per block and each $D_i \in \{A, S\}$. Only the existence assertion is proved here, which is all that Theorem 3.11 consumes; see Remark 3.9.
Def. 3.10.3	Definition 4.2	None.
Lem. 3.10.4	Lemma 2.7	Stated at an arbitrary coordinate i rather than at the first, and with the Taylor identity relativized to D.
Thm. 3.10.5	Theorem 5.1, Theorem 5.2	Split, and *stronger* in hypotheses. The source proves the enlargement step inside the proof of the absorption statement; here it is a separate theorem, and the iteration is an explicit induction on the star exponent. The source also assumes the left center is nontrivial; that hypothesis is dropped here, the degenerate cases $C = \varnothing$ and $C = A$ being covered by Remark 4.4.
Thm. 3.10.6	Theorem 6.1	None.
Def. 3.10.7	Definition 6.2	None.
Cor. 3.10.8	Corollary 6.3	None.
Lem. 3.10.9	Lemma 7.1	The source notes that S is symmetric; that is not used, here or there, and is recorded only as an observation.
Cor. 3.10.10	Corollary 8.1	The source iterates the doubling trick to arities $2 + 2^k$; here a maximal arity is taken and only the strict increase $2M - 2 > M$ is needed.
Cors. 3.10.11–3.10.13	—	Not covered; see Remark 8.3.
Props. 3.10.14–3.10.19	—	Not covered; see Remark 8.3.

The numbering of §3.8 follows the printed notes; Proposition 3.8.2 is the statement labelled `prop-essential-down` in the source, Proposition 3.8.3 is `prop-absorption-essential`, Theorem 3.8.5 is `absorption-essential` and Lemma 3.8.6 is `essential-groups`.

References

[1] L. Barto and M. Kozik, *Absorbing subalgebras, cyclic terms, and the constraint satisfaction problem*, Logical Methods in Computer Science **8** (2012), no. 1, 1–26.

[2] L. Barto and A. Kazda, *Deciding absorption*, International Journal of Algebra and Computation **26** (2016), no. 5, 1033–1060.

[3] Z. Brady, *Notes on CSPs and polymorphisms*, arXiv:2210.07383; current version at `https://notzeb.com/csp-notes.pdf`.

[4] D. Zhuk, *A proof of the CSP dichotomy conjecture*, Journal of the ACM **67** (2020), no. 5, article 30.